

TASK 1: STUDENT DATABASE (studentDB)

1.Create Database: use studentDB

```
mongosh mongodb://127.0.0.1:27017/ x + v
Microsoft Windows [Version 10.0.26200.8328]
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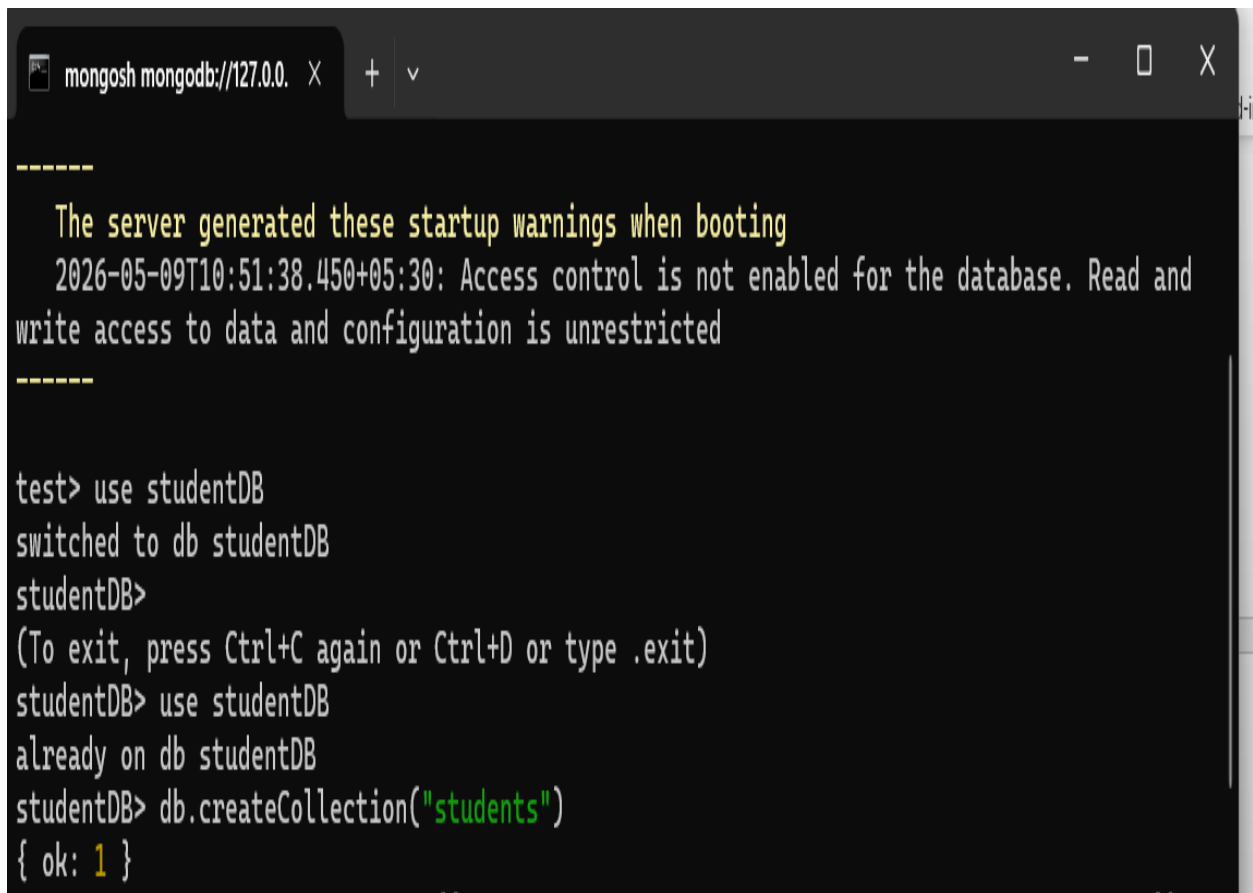
C:\Users\Kanagaraj>mongosh
Current Mongosh Log ID: 6a0086a2599c2dbfcbabc113
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=
mongosh+2.8.3
Using MongoDB:      8.2.7
Using Mongosh:      2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-05-09T10:51:38.450+05:30: Access control is not enabled for the database. Read and write access to data
  and configuration is unrestricted
-----

test> |
```

2.Create Collection: students



```
mongosh mongodb://127.0.0.1 X + v - □ X
-----
The server generated these startup warnings when booting
2026-05-09T10:51:38.450+05:30: Access control is not enabled for the database. Read and
write access to data and configuration is unrestricted
-----

test> use studentDB
switched to db studentDB
studentDB>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
studentDB> use studentDB
already on db studentDB
studentDB> db.createCollection("students")
{ ok: 1 }
```

3.Insert One and Many Documents

```
mongosh mongodb://127.0.0.1 X + v
Using MongoDB: 8.2.7
Using Mongosh: 2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
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-----

test> use studentDB
switched to db studentDB
studentDB>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
studentDB> use studentDB
already on db studentDB
studentDB> db.createCollection("students")
{ ok: 1 }
studentDB> db.students.insertOne({ name: "Ravi", age: 20, department: "CSE", marks: 85 })
{
  acknowledged: true,
  insertedId: ObjectId('6a008750599c2dbfcbabc114')
}
studentDB> db.students.insertMany([
| { name: "Anu", age: 19, department: "ECE", marks: 92 },
| { name: "Kumar", age: 21, department: "MECH", marks: 76 },
| { name: "Priya", age: 20, department: "CSE", marks: 88 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a008769599c2dbfcbabc115'),
    '1': ObjectId('6a008769599c2dbfcbabc116'),
    '2': ObjectId('6a008769599c2dbfcbabc117')
  }
}
```

4.Read Operations (find, filter)

```
mongosh mongodb://127.0.0.1:27020/
studentDB> db.students.find()
[
  {
    _id: ObjectId('6a008750599c2dbfcbabc114'),
    name: 'Ravi',
    age: 28,
    department: 'CSE',
    marks: 85
  },
  {
    _id: ObjectId('6a008769599c2dbfcbabc115'),
    name: 'Anu',
    age: 19,
    department: 'ECE',
    marks: 92
  },
  {
    _id: ObjectId('6a008769599c2dbfcbabc116'),
    name: 'Kumar',
    age: 21,
    department: 'MECH',
    marks: 76
  },
  {
    _id: ObjectId('6a008769599c2dbfcbabc117'),
    name: 'Priya',
    age: 20,
    department: 'CSE',
    marks: 88
  }
]
```

```
mongosh mongodb://127.0.0.1 X + v
}
]
studentDB> db.students.find({ department: "CSE" })
[
  {
    _id: ObjectId('6a008750599c2dbfcbabc114'),
    name: 'Ravi',
    age: 20,
    department: 'CSE',
    marks: 85
  },
  {
    _id: ObjectId('6a008769599c2dbfcbabc117'),
    name: 'Priya',
    age: 20,
    department: 'CSE',
    marks: 88
  }
]
```

5. Update Operations (\$set, \$gt)

```
]
]
studentDB> db.students.find({ department: "CSE" })
[
  {
    _id: ObjectId('6a008750599c2dbfcbabc114'),
    name: 'Ravi',
    age: 20,
    department: 'CSE',
    marks: 85
  },
  {
    _id: ObjectId('6a008769599c2dbfcbabc117'),
    name: 'Priya',
    age: 20,
    department: 'CSE',
    marks: 88
  }
]
studentDB> db.students.updateOne({ name: "Ravi" }, { $set: { marks: 90 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.deleteOne({ name: "Kumar" })
{ acknowledged: true, deletedCount: 1 }
studentDB> |
```